SHRESTHANSH NAYAK

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PROFILE

Undergraduate Physics student with a strong interest in theoretical and mathematical physics, computational science, and astronomy. Interested in understanding physical systems through mathematical reasoning, programming, and independent study.

EDUCATION

B.Sc. (Hons.) in Physics — S.K.C.G. Autonomous College, Paralakhemundi | 2025–2028 (Pursuing)

Class XII — Centurion Public School | 86%

Class X — New Life English Medium School | 90%

ACADEMIC INTERESTS

Theoretical Physics · Mathematical Physics · Quantum Mechanics · Special and General Relativity · High Energy and Particle Physics · Astronomy and Cosmology · Computational Physics

TECHNICAL SKILLS

Programming: Python, Java, Numpy ,SQL

Foundations: Calculus, Linear Algebra, Differential Equations, Mathematical Methods, Classical Mechanics, and Modern Physics

General: Problem solving, mathematical analysis, and scientific reasoning

SELECTED PROJECTS

Pendulum Simulation

Developed a Python-based numerical simulation to explore pendulum motion and its dynamics. The work also includes a double-pendulum simulation using the fourth-order Runge–Kutta (RK4) method to study coupled, nonlinear motion.

Projectile Motion Simulator

Built a Python simulation of projectile motion, including the effect of air resistance, to explore how forces influence trajectories.

INDEPENDENT RESEARCH & EXPLORATION

Explored some basic mathematical conjectures such as Riemann Hypothesis, Collatz conjecture.

Had recent interest on Little Red Dots (LRD) spotted by James Web Space Telescope and Blackhole Sun.

These are independent learning and exploration activities, not peer-reviewed publications.

HOBBIES & INTERESTS

Music · Movies · Online Games · Badminton · Writing

LANGUAGES

English · Hindi · Odia